

// Day 4 of 10

Day 4

10 Java Tips

Developers make these every day.
Don't be one of them.

→ Swipe through all 10



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Mistake #01

Use Optional Properly

Stop returning null — causes NPEs

```
// Bad
public User findUser(Long id){
    return null; // caller forgets check
}
```

```
// Good
public Optional findUser(Long id){
    return repo.findById(id);
}
```



Optional forces callers to handle absence



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Mistake #02

Use var for Local Variables

Reduces boilerplate, cleaner code

```
// Bad
HashMap> map
    = new HashMap<>;

// Good
var map = new HashMap<>;
var users = repo.findAll();
// Java 10+ feature
```



var works great for local variables



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Mistake #03

Use Streams Correctly

Cleaner than for-each loops

```
// Bad
List names = new ArrayList<>();
for(User u : users){
    if(u.isActive())
        names.add(u.getName());
}
```

```
// Good
var names = users.stream()
```



Streams = readable, functional style



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Mistake #04

StringBuilder for Concatenation

+ in loops creates many objects

```
// Bad
String s = "";
for(int i = 0; i < 1000; i++)
    s += i; // new object each time!
```

```
// Good
var sb = new StringBuilder();
for(int i = 0; i < 1000; i++)
    sb.append(i);
```



Use StringBuilder in loop concatenation



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Mistake #05

Records for Data Classes

Replaces verbose POJOs/DTOs

```
// Bad
public class UserDTO {
    private String name;
    private String email;
    // + getters, setters, equals ...
}
```

```
// Good
public record UserDTO(
```



Records auto-generate everything



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Mistake #06

Switch Expressions

Cleaner than old switch statements

```
// Bad
String label;
switch(status){
    case 1: label="Active"; break;
    case 2: label="Inactive"; break;
    default: label="Unknown";
}
```

```
// Good
```



Switch expressions — Java 14+



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Mistake #07

Immutable Collections

Prevent accidental modification

// Bad

```
List list = new ArrayList<>();  
list.add("a");  
// Anyone can modify this!
```

// Good

```
List list = List.of("a","b","c");  
Map map = Map.of("a",1);  
// Immutable – safe to share
```



List.of() / Map.of() since Java 9



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Mistake #08

Try-with-Resources

Always close your resources

```
// Bad
```

```
FileReader fr = new FileReader("f.txt");  
// Exception? fr never closed!
```

```
// Good
```

```
try(var fr = new FileReader("f.txt")){  
    // fr auto-closed after block  
}
```



Implements AutoCloseable = use try-with



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Mistake #09

Avoid Raw Types

Raw types skip compile-time checks

// Bad

```
List list = new ArrayList();  
list.add("text");  
list.add(123); // No compile error!
```

// Good

```
List list = new ArrayList<>();  
list.add("text");  
// list.add(123); // Compile error!
```



Always use generics — no raw types



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Mistake #10

Comparator.comparing

Cleaner sorting for objects

```
// Bad
users.sort((a,b) → {
  return a.getName().compareTo(b.getName());
});
```

```
// Good
users.sort(
  Comparator.comparing(User::getName)
    .thenComparing(User::getAge)
```

💡 Chain comparators cleanly



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done = true

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